

# Finite Symbolic Dynamics

Function, Resolution, and the Terminus of Symbolic Growth

Part III: Toward a Stable Operational Grammar for Finite Dynamics

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# Prefatory Note

The first two parts of *Finite Symbolic Dynamics* established a developing framework rather than a completed formal system. Part I moved from nonlinear dynamics toward an endogenous dynamics of finite symbolic trajectories. It gave priority to registered sequence over constructed geometry, treated the analytical process as another symbolic trajectory, and retained uncertainty, provenance, transformation history, and admissibility as part of the result. Part II moved upstream. It examined the transition from sensing to symbol, the Generonic boundary, the role of the Transfector, finite instrument resolution, and the recursive coupling between measurement, language, instrumentation, and symbolic transformation.

The present third part arose because the development itself became observable. As the language of FSD expanded, each new distinction could be locally justified. Yet the continued addition of terms, partitions, qualifications, and specialised symbolic objects also revealed a danger. A finite symbolic framework may still produce an indefinitely growing explanatory trajectory. The Alphonic limit places a terminus upon the resolution of first-order distinction, but it does not by itself place a terminus upon the length, branching, or lexical density of the symbolic trajectory constructed above that boundary.

This observation turns the developing framework back upon itself. The conversations, papers, revisions, diagrams, and formal proposals that produced FSD can themselves be treated as a recorded symbolic trajectory. They exhibit episodes of expansion, compression, correction, recurrence, and local instability. During the present development, the framework approached a region in which further local partitioning threatened to increase symbolic complexity faster than functional value. The resulting correction did not require another subdivision. It required a return to an existing word whose significance had not yet been sufficiently developed: *functional*.

The central question of Part III is therefore not simply what a Functional Symbolic Trajectory is made from. It is:

What makes a symbolic trajectory functional, and what permits it to stop growing?

 (1)

This question introduces a second kind of terminus. The Alphonic terminus marks a limit of available distinction. A functional terminus marks a local condition in which further symbolic elaboration is presently unnecessary for the declared task. The first is a hard boundary of admissible distinction. The second is elastic. It can be reopened by new measurement, altered operating conditions, or failure of function, but it is not reopened merely because another symbolic subdivision can be imagined.

The numerical chapters in this monograph provide the first controlled setting in which this principle is examined. Positional arithmetic is normally written through a finite base while permitting an indefinitely extensible sequence of positions. Within FSM, this separation must be made explicit. A base does not itself define numerical resolution. Resolution depends upon the base, the available numerical character depth, the represented scale, and the finite registration

process by which the numerical state is realised. Ink marks, beans, abacus beads, mechanical counters, and computational shift registers are materially different, yet they can instantiate functionally equivalent finite counting and shifting operations. Arithmetic can therefore be examined as a finite symbolic trajectory of distinguishable state transitions rather than as manipulation of an implicitly infinitely resolved numerical continuum.

This monograph remains deliberately provisional. Its purpose is not to close the terminology of FSD, but to make the conditions of useful closure visible. In particular, the terms Nexil and Dynexil are not extended merely to fill newly noticed conceptual space. The Dynexil retains its present local role as a bounded delay-structured dynamical descriptor. Numerical representation is developed first through existing FSM commitments and only then examined for whether additional terminology is warranted.

The final chapters establish conditions for exposing the resulting grammar to a difficult external application. The intended next study is a finite FSM treatment of the Navier–Stokes problem. That programme should not begin by expanding the language until the application fits. It should begin with the compact grammar developed here and allow the application to test whether that grammar is functionally sufficient.

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## Part I

# Function and the Dynamics of Symbolic Growth

# Chapter 1

## The Problem of Unbounded Symbolic Growth

### 1.1 Finite Symbols Do Not Imply Finite Explanations

Finite Symbolic Mechanics begins from finite distinction. A registered symbol has finite extent, finite resolution, finite provenance, and finite conditions under which it can be distinguished. This commitment prevents the first-order symbol from being treated as a zero-extent ideal object carrying unlimited resolution.

It does not, however, follow that a sequence constructed from finite symbols must itself possess a naturally imposed terminal length. Let

$$S_i \in \mathcal{S}_{\text{finite}} \quad (1.1)$$

for every registered symbolic state. A trajectory may nevertheless take the form

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \cdots \rightarrow S_n \rightarrow \cdots . \quad (1.2)$$

Each member of the sequence is finite. The potential continuation of the sequence is a different problem.

The same distinction appears in scientific language. A framework may contain only finite terms, equations, diagrams, and definitions while continuing to create new subcategories whenever a new local difficulty appears. The finitude of each symbol therefore does not guarantee the functional economy of the symbolic system.

This gives two different forms of recursive extension. The first is refinement of distinction:

$$\Delta x, \quad \frac{\Delta x}{2}, \quad \frac{\Delta x}{4}, \quad \dots \quad (1.3)$$

until the Alphonic limit is reached. The second is elaboration of representation:

$$\Gamma_0 \rightarrow \Gamma_1 \rightarrow \Gamma_2 \rightarrow \cdots , \quad (1.4)$$

where each  $\Gamma_i$  is a symbolic construction, explanation, model, or trajectory derived from the earlier state.

The first problem already has a declared FSM terminus:

$$\Delta x \geq \alpha. \quad (1.5)$$

The second does not.

## 1.2 The Risk of Recursive Partition

Partitioning is one of the most useful operations in science. A broad distinction is divided into finer distinctions; the finer distinctions expose differences; the differences permit additional measurement, classification, or control. Part II described this process as an important source of increasing symbolic resolution.

The difficulty appears when partitioning becomes its own justification. If every new local difference automatically earns a new named category, the symbolic framework can become a recursively articulated tree whose growth is limited principally by available time, storage, and attention. Such a tree may still contain valid local statements. The concern is not that its branches are necessarily wrong. The concern is whether the growing branch structure continues to perform proportionate functional work.

A framework can therefore enter a region in which

$$\Delta\text{symbolic structure} \gg \Delta\text{functional gain.} \quad (1.6)$$

This condition is not yet a formal instability criterion. It is an observational warning. It indicates that the governing commitments should be examined before another local partition is introduced.

## 1.3 Why the Problem Is Internal to FSD

Finite Symbolic Dynamics cannot place itself outside the symbolic domain that it analyses. If exposition is itself part of the trajectory, then the growth of FSD terminology is a legitimate object of FSD inspection. The framework must therefore be capable of asking not only whether a new term is coherent, but also what effect the term has upon the larger trajectory of explanation.

This gives a reflexive requirement:

$$\boxed{\text{The growth of the descriptive FST is itself a dynamical observable.}} \quad (1.7)$$

The statement is intentionally modest. It does not imply that every conceptual change possesses a simple numerical measure. It says only that the sequence of recorded changes has order, history, local neighbourhood, recurrence, and transformation structure. Those are precisely the features that FSD is intended to preserve.

## Chapter 2

# The Container Functional Symbolic Trajectory

### 2.1 A Research Programme as a Trajectory

A long research programme can be represented schematically as an accumulating trajectory

$$\Gamma_C = \{S_0, T_1, S_1, T_2, S_2, \dots, T_n, S_n\}, \quad (2.1)$$

where the subscript  $C$  denotes a *container* trajectory. The term is descriptive rather than foundational. It marks the fact that many smaller Functional Symbolic Trajectories can be held within a wider historical sequence.

A paper, monograph, diagram, equation, or definition can then be understood as a temporary stabilisation or compression of a local region of this larger trajectory. If  $C$  is a compression operation, a document may be written schematically as

$$D_j = C\left(\Gamma_C^{[a:b]}\right). \quad (2.2)$$

The document is useful because it allows a complex trajectory to be handled as a compact object. Yet

$$D_j \neq \Gamma_C^{[a:b]}. \quad (2.3)$$

The compression does not erase the existence of the path from which it was produced.

### 2.2 Documents as Local Stabilisations

This interpretation changes the status of a completed document. Publication does not necessarily mean that a global conceptual trajectory has ended. It means that a local region has been stabilised sufficiently to be communicated, tested, and later reopened if necessary.

A subsequent investigation may read the document as a compact symbolic input:

$$D_j \rightarrow \Gamma_{j+1}. \quad (2.4)$$

The earlier compression thereby becomes a future perturbation. Scientific writing participates in the dynamics rather than merely describing it from outside.

This gives a useful distinction between *global openness* and *local closure*. The global container FST may continue to grow. A local functional trajectory can nevertheless be considered complete

for a declared purpose.

## 2.3 Reflexive Observation Without Infinite Self-Description

There is an obvious danger in making the framework reflexive: every description of the framework could generate another description of the description. The purpose of the container FST is not to create a new recursive level each time reflection occurs. It is to provide a single operational reminder that the current exposition has a history.

The principle of economy therefore applies immediately. Reflection is warranted when it changes function. It is not warranted merely because another meta-level can be named.

## Chapter 3

# Recovering the Functional in Functional Symbolic Trajectory

### 3.1 From Trajectory to Function

The terms *symbolic* and *trajectory* have so far carried much of the explanatory load. A Functional Symbolic Trajectory preserves finite symbolic states and the ordered path through which they are transformed. Part III now gives explicit attention to the first word.

A trajectory is functional when its finite symbolic organisation supports a repeatable and operationally useful relation among registered states, transformations, and, where applicable, exogenous measurement. The relevant function must be declared rather than assumed.

Let  $M$  denote a set of exogenous registrations and let  $\Gamma$  denote a symbolic trajectory. A provisional functional relation can be written

$$M \xleftrightarrow{\Gamma} R, \quad (3.1)$$

where  $R$  is a repeatable registered result produced under declared conditions.

The function may be predictive, discriminative, reconstructive, comparative, communicative, or controlling. Nothing in the present definition requires the trajectory to reveal an inaccessible global state. Its adequacy is judged by what it does under stated conditions.

### 3.2 Functional Adequacy

Suppose a trajectory  $\Gamma$  maps a present registration toward a prediction of a later registration:

$$M_t \xrightarrow{\Gamma} \widehat{M}_{t+1}. \quad (3.2)$$

A later exogenous measurement produces

$$M_{t+1}. \quad (3.3)$$

Repeated comparison of  $\widehat{M}_{t+1}$  and  $M_{t+1}$  provides one possible form of functional assessment.

In other domains, function may mean that a symbolic procedure reliably separates two experimentally distinguishable classes, reconstructs a previously withheld registered sequence, or permits a device to maintain control within a measured range. Function is therefore tied to operation and consequence rather than to the mere richness of description.

A longer trajectory is not automatically more functional. If

$$|\Gamma_2| > |\Gamma_1|, \quad (3.4)$$

we cannot infer

$$\mathcal{F}(\Gamma_2) > \mathcal{F}(\Gamma_1). \quad (3.5)$$

The relationship between complexity and function must be examined.

### 3.3 A Standard Model as a Stable Container FST

A successful scientific framework may be treated, in this language, as a relatively stable container FST from which many local trajectories can be generated without continually inventing new first principles. Its importance lies not only in the number of observations it can describe but in its ability to relate them through a compact and reusable grammar.

This does not require a claim that the symbolic framework is identical to the measured world. It means that the framework provides a stable relational system through which many exogenous registrations can be compared and reproduced.

The present programme therefore aims not at an unlimited taxonomy of dynamics but at a compact operational grammar from which many useful trajectories can be formed. In this sense, Part III begins to explore the possibility of a functional *standard model of symbolic dynamics*: not a final ontology, but a stable and economical set of operations adequate across multiple domains.

## Chapter 4

# Alphonic Terminus and Functional Terminus

### 4.1 Two Different Stopping Conditions

The Alphonic limit and the proposed functional terminus address different problems.

The Alphonic terminus says that further first-order distinction is unavailable within the declared measurement system. Schematically,

$$\cdots \rightarrow 4\alpha \rightarrow 2\alpha \rightarrow \alpha \rightarrow \boxed{\text{no finer admissible distinction}}. \quad (4.1)$$

The functional terminus says that further symbolic distinction is presently unnecessary for the declared task. Let  $\Gamma^*$  denote a locally sufficient functional trajectory. Then

$$\Gamma \rightarrow \Gamma^* \quad (4.2)$$

may terminate operationally even though further symbolic continuations could be imagined.

The distinction can be written compactly as

|                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Alphonic terminus: further registered distinction is unavailable,<br>Functional terminus: further symbolic distinction is presently unnecessary. |
|--------------------------------------------------------------------------------------------------------------------------------------------------|

(4.3)

### 4.2 The Elastic Character of Functional Closure

A functional terminus is not final closure. It is local and elastic. A new measurement can perturb the trajectory:

$$M_{\text{new}} : \Gamma^* \rightarrow \Gamma'^*. \quad (4.4)$$

A changed operating range can do the same. So can loss of prediction, a failure of reconstruction, or the discovery that two previously merged states are experimentally distinguishable.

What does not by itself reopen the trajectory is the fact that another conceptual subdivision can be imagined. This gives a central working rule:

|                                                                              |
|------------------------------------------------------------------------------|
| The possibility of making a distinction is not sufficient reason to make it. |
|------------------------------------------------------------------------------|

(4.5)

A distinction earns its place by performing functional work.

### 4.3 Local Closure Without Global Completion

The difference between local and global closure is important because FSM does not require a claim of globally completed knowledge. A local model may be sufficient within an operating region:

$$\Gamma_{\Omega}^* \tag{4.6}$$

for declared domain  $\Omega$ . Outside  $\Omega$ , the trajectory may fail or require extension.

This allows scientific work to stop without pretending that inquiry itself has ended. It also permits a monograph, model, or numerical representation to be functionally complete while remaining historically open.

## Chapter 5

# Symbolic Elasticity and the Epicycle Problem

### 5.1 Local Improvement and Global Strain

The historical image of epicyclic astronomy is useful here if treated as a methodological analogy rather than as a simple story of error. Additional local machinery can improve agreement with observation while simultaneously increasing the structural burden carried by the framework.

A symbolic system may therefore show

$$\Delta\mathcal{C} > 0 \tag{5.1}$$

while also showing

$$\Delta\mathcal{F} > 0. \tag{5.2}$$

This is not necessarily a problem. Early extensions may be highly productive. The warning appears when increasingly large symbolic additions produce increasingly small functional returns.

For conceptual purposes only, define

$$\eta_F = \frac{\Delta\mathcal{F}}{\Delta\mathcal{C}}. \tag{5.3}$$

Neither  $\mathcal{F}$  nor  $\mathcal{C}$  is assumed here to possess a universal scalar definition. The expression is a diagnostic relation. It asks whether the trajectory is obtaining useful functional gain from the complexity it adds.

### 5.2 Epicyclic Growth

A region of possible epicyclic growth may be characterised schematically by

$$\Delta\mathcal{C} \gg 0, \quad \Delta\mathcal{F} \approx 0. \tag{5.4}$$

The appropriate response need not be another correction. It may be reconsideration of a generating commitment.

Suppose framework  $A$  uses commitments  $C_A$  and requires a growing set of auxiliary corrections

$$C_A + e_1 + e_2 + e_3 + \cdots. \tag{5.5}$$

A changed commitment  $C_B$  may produce a different family of trajectories without the same correction burden. If

$$\mathcal{C}(\Gamma_B) < \mathcal{C}(\Gamma_A) \quad (5.6)$$

and

$$\mathcal{F}(\Gamma_B; M) \geq \mathcal{F}(\Gamma_A; M), \quad (5.7)$$

then the reformulation has achieved functional compression.

The criterion is not aesthetic simplicity. It is measured or operational economy under comparable function.

### 5.3 A Warning for the Development of FSM

The same discipline must apply to FSM and FSD. A new term may be locally clear yet globally expensive. The framework should therefore ask whether a new noun reduces many future trajectories or merely names one local difficulty.

A term that repeatedly shortens, clarifies, or preserves multiple functional trajectories may become stable. A term that requires repeated redefinition across applications may be evidence that the underlying partition is not carrying enough functional value.

This principle is particularly relevant to the developing use of the Dynexil. Its present function is local and specific: it preserves a delay-structured symbolic neighbourhood. Whether that role transfers successfully across quantum measurement, language, numerical sequences, and fluid dynamics is an empirical question for the research programme, not a reason to broaden the definition in advance.

## Chapter 6

# Stable Basins and the Dynamics of Framework Construction

### 6.1 A Recurrent Conceptual Basin

The development of FSM and FSD has undergone repeated perturbation. New questions have introduced new vocabulary, new mathematical forms, and new applications. Yet the trajectory repeatedly returns toward a relatively compact group of commitments:

{finite distinction, measurement, trajectory, uncertainty, provenance, admissibility, function}.  
(6.1)

It is useful to describe this as a recurrent conceptual basin. The phrase is deliberately operational. It does not assert that a literal ontological attractor has been proved. It states that repeated symbolic perturbations appear to return the construction toward a stable region of relational commitments.

### 6.2 Attractor-Like Stability as a Constructed Observation

Part I treated phase geometry as a constructed representation. The same caution applies here. If a coded history of the framework were embedded, projected, or otherwise analysed and a stable region appeared, the resulting geometry would remain a product of the stated transformation.

The scientifically useful question is therefore not

What is the hidden attractor of the theory?

but

Which commitments recur and remain functionally stable across different perturbations and different admissible reconstructions of the recorded trajectory?

If the same compact grammar survives repeated applications, the recurrence becomes evidence for functional stability. If each new application requires an expanding specialised vocabulary, the growth becomes evidence of strain.

### 6.3 Growth as Measurement

This gives a possible practical monitoring procedure for the research programme. At major revisions, record:

- which terms were introduced or removed;
- which existing operations changed meaning;
- what new exogenous or computational function was gained;
- whether old exceptions disappeared or multiplied;
- whether the new structure transferred to more than one domain.

The list is not intended as bureaucracy. It provides a minimal provenance trace through which symbolic growth can be distinguished from functional improvement.

## **Part II**

# **Finite Number Representation as a Functional Test Case**

## Chapter 7

# Numerical Representation as a Functional Stress Test

### 7.1 Counting Before Number Space

Arithmetic provides a particularly useful test of the emerging framework because counting can be implemented through visibly finite physical interactions. Five beans, five marks of ink, five notches, five abacus beads, five mechanical counter states, or five registered electronic transitions can all participate in a stable counting relation.

The material realisations differ. Their functional identity can be preserved.

Let  $B_i$  denote a distinguishable bead, bean, mark, or registered unit. A simple count may be represented as

$$\{B_1, B_2, \dots, B_n\} \longrightarrow n. \quad (7.1)$$

The numeral  $n$  is not identical to the collection. It is a finite symbolic registration produced under a counting rule.

This makes arithmetic naturally compatible with the FSD emphasis on operations. The primitive explanatory language becomes

$$\text{distinguish} \rightarrow \text{count} \rightarrow \text{group} \rightarrow \text{shift} \rightarrow \text{register}. \quad (7.2)$$

The numerical noun is the compressed endpoint of an operational trajectory.

### 7.2 Physical Interactional Identity

For counting to function, each counted event or object must be distinguishable under the relevant rule. The identity need not be metaphysically permanent. It needs to be stable enough for the counting operation.

A bead moved from one side of an abacus to another is a physical interaction. A new ink mark changes the page. A bean transferred between containers changes a finite arrangement. A bit pattern changes through switching events in a computational register. The symbolic relation survives because each substrate supplies repeatable distinctions and ordered transformation rules.

This suggests a substrate-independent functional form:

$$\mathfrak{I}_n \xrightarrow{\mathcal{O}} \mathfrak{I}_{n+1}, \quad (7.3)$$

where  $\mathfrak{I}_n$  is a finite interactional identity state and  $\mathcal{O}$  is a declared successor operation.

The abstraction is downstream of the physical realisation. It is justified by repeated functional equivalence across different realisations.

## Chapter 8

# Base Is Not Resolution

### 8.1 The Base as Alphabet and Carry Relation

A positional base supplies a finite alphabet and a positional rule. For base  $b$ , let

$$\mathcal{A}_b = \{0, 1, \dots, b-1\}. \quad (8.1)$$

The base determines how many distinct digit states are available at a single numerical character position and when a carry or shift to the next position occurs.

The base alone does not determine the numerical resolution of a finite representation. A base-10 numeral can carry two significant digit positions, nine positions, or one hundred positions. Treating all of these as members of the same base does not make them equal in resolving power.

The classical symbolic grammar of positional notation commonly permits further positions to be appended without a formal terminal character depth. Within pure mathematics this is a legitimate construction. Within FSM, difficulty arises when indefinite symbolic extension is allowed to inherit the status of measurement-supported resolution.

The present argument therefore does not reject positional base. It rejects the silent equivalence

$$\text{fixed base} + \text{indefinite character extension} \equiv \text{finite measured number}. \quad (8.2)$$

### 8.2 Base, Character Depth, and Scale

Let a finite positional register be written

$$R_{b,m,e} = (d_0, d_1, \dots, d_{m-1}; b, m, e), \quad (8.3)$$

where

- $b$  is the numerical base;
- $m$  is the available numerical character depth;
- $e$  identifies the scale of the most significant character position;
- each  $d_j \in \mathcal{A}_b$ .

If  $d_0$  occupies scale  $b^e$ , the represented value is

$$\mathcal{V}(R_{b,m,e}) = \sum_{j=0}^{m-1} d_j b^{e-j}. \quad (8.4)$$

The smallest adjacent positional distinction available within that register is therefore

$$\Delta_{b,m,e} = b^{e-m+1}. \quad (8.5)$$

This relation makes the central point explicit:

$$\text{numerical resolution} = f(\text{base}, \text{character depth}, \text{scale}). \quad (8.6)$$

The base is not itself the resolution.

### 8.3 The Working Base-10 Nine-Character Register

For the computational investigations proposed here, a working convention is adopted:

$$b = 10, \quad m = 9. \quad (8.7)$$

The nine positions are treated as nine significant numerical character spaces. Sign, radix marker, and scale label are structural metadata rather than additional significant digit positions. This convention is explicit and may later be revised if the experimental programme gives reason to change it.

For a most significant decimal scale  $10^e$ , the least available positional distinction is

$$\Delta_{10,9,e} = 10^{e-8}. \quad (8.8)$$

Thus

$$1.23456789 \times 10^{20} \quad (8.9)$$

and

$$1.23456789 \times 10^{-20} \quad (8.10)$$

use the same nine-character significand depth while possessing very different absolute resolution.

The scale may move. The admitted character depth remains finite.

### 8.4 Resolution Per Character Space

The relation above can be interpreted as a resolution per available character space. Each additional position refines the represented scale by a factor of  $b$ . If the character depth is fixed, the numerical register does not gain unlimited resolution merely by existing within a base whose grammar permits further positions.

This distinction is fundamental for measurement. A value written with nine supported character positions does not acquire ten, twenty, or infinitely many measurement-supported positions because later algebra can generate them. Additional characters are additional symbolic structure whose provenance must be declared.

## Chapter 9

# Numerical Alphons, Nexils, and Higher-Order Representation

### 9.1 Why Terminology Must Be Introduced Cautiously

The numerical case invites new terms, and therefore provides an immediate test of the functional discipline developed in Part I of this monograph. The existing concepts should be used before new entities are created.

The Alphonic principle already says that a symbol has a lower bound of distinguishability within a declared symbolic system. A numerical character position can therefore be treated as a local Alphonic context without requiring that a new foundational particle of arithmetic be invented.

Likewise, the Nexil already denotes a first-order finite symbolic registration generated from measurement at the relevant boundary of distinction. It should not be stretched automatically to mean every downstream digit, register, intermediate arithmetic value, or computed state.

### 9.2 A Numerical Alphon as a Character-Space Distinction

A useful provisional language is to describe an admissible digit-state at one positional character space as a *numerical Alphonic distinction*. Let

$$a_j^{(b)} \in \mathcal{A}_b \tag{9.1}$$

denote the registered digit state at character position  $j$ .

The term *numerical Alphon* may be used as shorthand for this smallest declared character-space distinction if, and only if, the later applications show that the abbreviation performs useful repeated work. The present monograph does not require it as a new primitive.

The important structure is operational:

$$a_j^{(b)} \xrightarrow{\text{admissible change}} a_j'^{(b)}. \tag{9.2}$$

The physical realisation may be ink, bead position, mechanical state, or encoded register state.

### 9.3 Higher-Order Numerical Representation

A complete numeral is not simply one digit enlarged. It is a relational assembly of digit states occupying ordered positions. In the language of Part I, this is a higher-order symbolic relation, not a higher physical dimension.

Define

$$\mathcal{R}_{b,m,e} = [a_0^{(b)}, a_1^{(b)}, \dots, a_{m-1}^{(b)}]. \quad (9.3)$$

The order of the representation records simultaneous positional relations. Its numerical meaning depends upon the base and scale mapping

$$\mathcal{V} : \mathcal{R}_{b,m,e} \rightarrow \text{finite registered value.} \quad (9.4)$$

The representation may therefore be called a higher-order numerical symbolic object without implying an additional measured dimension.

### 9.4 When the Whole Numeral Is Measurement-Derived

Suppose an exogenous measurement  $M_t$  crosses the Generonic boundary and is stabilised as a Nexil

$$N_\alpha(M_t). \quad (9.5)$$

A numerical representation of that Nexil may then be constructed:

$$N_\alpha(M_t) \xrightarrow{\mathcal{R}_{b,m,e}} \mathcal{R}_{b,m,e}(N_\alpha). \quad (9.6)$$

The numerical register is therefore a representation of the measurement-derived symbolic state. It should not silently be treated as another measurement.

A useful full notation for bookkeeping may be

$$\mathcal{N}_t = (\mathcal{R}_{b,m,e}, \mathcal{U}_t, \mathcal{P}_t, \mathcal{A}_t), \quad (9.7)$$

where uncertainty, provenance, and admissibility remain attached.

Whether the entire object should eventually be called a *numerical Nexil* remains an open terminological question. If the name shortens many later trajectories without erasing the distinction between first-order measurement and endogenous representation, it may earn a place. If it creates ambiguity, the existing notation is sufficient.

## Chapter 10

# Symbolic Shift Operational Arithmetic

### 10.1 Arithmetic as Finite State Transition

Within the present frame, arithmetic is treated as a sequence of finite symbolic state changes under declared operational rules. Let

$$R_n = [d_0, d_1, \dots, d_{m-1}]_n \quad (10.1)$$

denote a finite numerical register. An arithmetic operation  $T$  acts as

$$R_n \xrightarrow{T} R_{n+1}. \quad (10.2)$$

The result is not assumed to exist first as an infinitely resolved number that the register later approximates. The admitted object of analysis is the finite transition itself.

### 10.2 Successor, Carry, and Shift

At a single base- $b$  position, a successor operation can be written

$$d_j \mapsto d_j + 1 \quad \text{if } d_j < b - 1. \quad (10.3)$$

At the upper digit state,

$$b - 1 \mapsto 0 \quad (10.4)$$

and a carry is transferred to the next admissible position.

Define a local shift/carry operator  $\sigma_j$  such that

$$\sigma_j : (d_j, d_{j-1}) \mapsto (0, d_{j-1} + 1) \quad (10.5)$$

when the lower position saturates. The exact indexing convention can vary with implementation; the important point is that the carry is an explicit symbolic operation rather than an invisible property of an abstract numeral.

A chain of such operations forms a Functional Symbolic Trajectory:

$$R_0 \xrightarrow{T_1} R_1 \xrightarrow{T_2} R_2 \xrightarrow{T_3} \dots \quad (10.6)$$

### 10.3 Beans, Ink, Beads, and Shift Registers

The same functional operation can be instantiated through different physical substrates. Consider four implementations of a base-10 counter.

An ink implementation draws one mark for each unit and groups or rewrites after ten marks. A bean implementation moves countable objects between containers and forms groups of ten. An abacus implementation shifts beads between stable positions. An electronic implementation changes finite register states under logical transition rules.

These systems are not materially identical. Their functional equivalence is demonstrated by a repeatable relation between states:

$$\phi_{AB}(R_n^{(A)}) = R_n^{(B)}, \quad (10.7)$$

where  $A$  and  $B$  denote two different substrates and  $\phi_{AB}$  preserves the declared counting relation.

This is a concrete example of the role of function. The symbolic operation is stabilised not because the substrates are the same but because the finite relational trajectory can be transferred reproducibly between them.

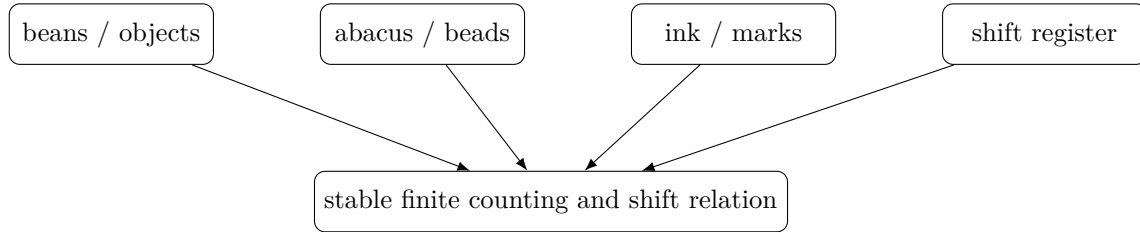


Figure 10.1: Different physical substrates can instantiate a functionally equivalent finite symbolic counting relation without being physically identical.

### 10.4 Operational Identity Rather Than Infinite Numerical Possession

The finite register does not need to possess an infinitely resolved value in order to function. It must preserve the relations required by the declared arithmetic operation at the admitted character depth and scale.

This distinction becomes especially important when arithmetic is used downstream of measurement. The operation may create new symbolic relations. It does not create new exogenous measurement depth merely by extending the written numeral.

## Chapter 11

# Carry, Overflow, Scale Shift, and Loss of Distinction

### 11.1 Finite Depth Creates Explicit Boundaries

A finite character depth makes conditions visible that indefinite notation can conceal. Consider the base-10, nine-character register

$$R = [9, 9, 9, 9, 9, 9, 9, 9, 9]. \quad (11.1)$$

Under the declared  $m = 9$  rule, the successor operation cannot silently append a tenth significant digit while preserving the same scale and resolution structure.

The system must instead perform a declared transition. Possible policies include scale shift, saturation, overflow signalling, or a change in admitted resolution. For a scale-shift policy,

$$9.99999999 \times 10^e \xrightarrow{+\Delta} 1.00000000 \times 10^{e+1}. \quad (11.2)$$

The shift preserves nine significant character positions but changes the absolute least distinction from

$$10^{e-8} \quad (11.3)$$

to

$$10^{e-7}. \quad (11.4)$$

Resolution has changed as a consequence of the scale transition.

### 11.2 Scale Shift Is Part of the Trajectory

In conventional notation a scale change can appear cosmetic because the represented mathematical quantity is treated as invariant under rewriting. Within FSM, the symbolic transformation is part of the result. The register before and after the shift has different absolute resolving structure.

Thus

$$R_{10,9,e} \xrightarrow{\text{scale shift}} R_{10,9,e+1} \quad (11.5)$$

is not merely a change of typography. It alters the smallest difference that the register can carry at that scale.

The operation and its provenance should therefore remain recoverable.

### 11.3 Loss of Distinction

Suppose two finite comparison states  $x$  and  $y$  are represented at a higher finite character depth  $m' > m$ . A declared projection to the working depth  $m$  may produce

$$\Pi_{m' \rightarrow m}(x) = \Pi_{m' \rightarrow m}(y). \quad (11.6)$$

The distinction between  $x$  and  $y$  has then been removed by the lower-depth representation.

This can be called a *symbolic collision*. The term describes an operational event rather than an error relative to an infinitely precise hidden number.

A finite comparison residue may be defined without invoking infinite precision:

$$\rho_{m,m'} = \mathcal{V}(R_{b,m',e}) - \mathcal{V}(\Pi_{m' \rightarrow m} R_{b,m',e}), \quad m' > m. \quad (11.7)$$

The residue is itself finite and depends upon the declared comparison depth  $m'$ . It records distinction removed during projection from one finite symbolic system to another.

### 11.4 Computed Digits Are Not Measured Digits

This yields a foundational numerical rule:

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| symbolically generated character depth does not retroactively increase measurement depth. |
|-------------------------------------------------------------------------------------------|

(11.8)

If a measurement supports nine numerical character positions and a downstream calculation prints twenty positions, the additional characters belong to an endogenous symbolic transformation unless independent measurement supports them.

The calculation may be useful. Its provenance must remain visible.

## Chapter 12

# Higher-Order Arithmetic and the Growth of Numerical FSTs

### 12.1 Composed Operations

Elementary operations become higher-order numerical trajectories when composed:

$$N_0 \xrightarrow{T_1} N_1 \xrightarrow{T_2} N_2 \xrightarrow{T_3} \dots \xrightarrow{T_n} N_n. \quad (12.1)$$

Each  $N_i$  is a finite registered numerical state under declared base, character depth, and scale conditions. Each  $T_i$  is an endogenous symbolic operation.

The resulting arithmetic object is therefore not merely  $N_n$ . The fuller object includes the transformation path:

$$\Gamma_N = \{N_0, T_1, N_1, T_2, \dots, T_n, N_n\}. \quad (12.2)$$

This is a numerical Functional Symbolic Trajectory.

### 12.2 Resolution Propagation

A calculation may preserve character depth while changing scale, or preserve scale while losing distinctions through projection. A useful state descriptor is therefore

$$N_i = (R_{b,m,e_i}, U_i, P_i, A_i). \quad (12.3)$$

The update is

$$N_{i+1} = T_i(N_i), \quad (12.4)$$

subject to an explicit registration rule

$$R_{b,m,e_i} \rightarrow R_{b,m,e_{i+1}}. \quad (12.5)$$

Resolution is part of the evolving state. It is not an invariant background resource.

### 12.3 Local Dynexil Analysis

The Dynexil remains a local trajectory descriptor. It is not the arithmetic update operator. Once a numerical FST has been formed, a local region may be examined through a delay-structured

bundle

$$D_t^{(k,\tau)} = [N_t, N_{t-\tau}, \dots, N_{t-k\tau}]_{U,P}. \quad (12.6)$$

This construction can be used to examine recurrence, local drift, symbolic collisions, scale transitions, or sensitivity to finite perturbation.

Whether the same local descriptor remains useful across numerical, linguistic, quantum, and fluid trajectories is precisely the kind of cross-domain functional test that can determine whether the term deserves long-term stability.

## 12.4 Finite Recurrence

A deterministic finite symbolic system with a finite accessible state set must eventually terminate, revisit a previous state, or enter a recurrent structure if allowed to run indefinitely. This fact is often described in conventional computation as a finite-precision artefact. Within FSD, the recurrence is instead part of the declared symbolic system and should be characterised rather than dismissed.

If

$$N_i = N_j, \quad i < j, \quad (12.7)$$

and the future update rule is deterministic and unchanged, then

$$N_{i+r} = N_{j+r} \quad (12.8)$$

for subsequent admissible steps  $r$ . The resulting cycle is a property of the finite symbolic dynamics under that rule, depth, and scale policy.

This becomes important in later nonlinear applications. Sensitive dependence should be examined in terms of distinguishable finite trajectories rather than infinitesimal separations that lie outside the declared registration system.

## **Part III**

# **Functional Economy and Readiness for External Application**

## Chapter 13

# Functional Economy as a Constraint on Language

### 13.1 Terms Must Perform Work

The numerical chapters demonstrate a wider principle. A new symbol or term should not be stabilised merely because a conceptual distinction can be stated. It should repeatedly reduce, preserve, or enable functional trajectories.

A provisional criterion is

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| A term earns stability when it compresses multiple functional trajectories without hiding distinctions required for function. |
|-------------------------------------------------------------------------------------------------------------------------------|

(13.1)

The final clause matters. Compression is indispensable, but excessive compression removes provenance and local dynamical structure. Insufficient compression produces an ever-expanding explanatory tree.

### 13.2 The Elastic Functional Region

The useful operating region lies between two failures:

$$\text{over-compression} \longleftrightarrow \boxed{\text{functional region}} \longleftrightarrow \text{over-elaboration.} \quad (13.2)$$

Over-compression replaces a trajectory with a noun that no longer carries distinctions needed for function. Over-elaboration preserves or invents more distinctions than the declared function requires.

The location of the functional region is not universal. It depends upon the problem, the available measurement, the desired operation, and the admitted uncertainty. This is why the boundary is elastic rather than Alphonic.

### 13.3 Vocabulary as a Dynamical Resource

Part II treated language as a resolution system. Part III adds a complementary constraint: increasing vocabulary can increase symbolic resolution while also increasing possible error structure and trajectory complexity.

A new term therefore changes not only the list of available symbols but the number of possible paths through the symbolic landscape. The appropriate question is not simply whether

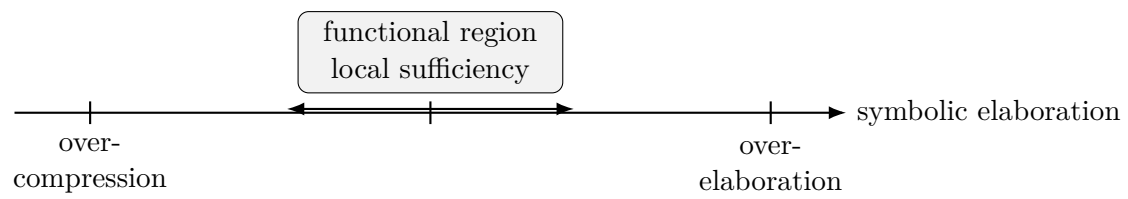


Figure 13.1: Functional sufficiency is treated as an elastic operating region rather than as an absolute point of closure.

the term is meaningful. It is whether the enlarged landscape improves stable function.

## Chapter 14

# A Provisional Functional Grammar of Finite Symbolic Dynamics

### 14.1 Operations Before Ontology

The aim of a functional grammar is to describe what the framework must do with the fewest stable operations presently required. A provisional sequence is

$$\boxed{\text{register} \rightarrow \text{relate} \rightarrow \text{transform} \rightarrow \text{retain provenance} \rightarrow \text{test function} \rightarrow \text{stop or continue.}} \quad (14.1)$$

This sequence does not replace the existing terminology. It constrains it.

### 14.2 The Present Minimal Vocabulary

The following terms presently appear to perform distinct repeated work.

**Finite Symbolic Mechanics.** FSM supplies foundational constraints concerning finite registration, uncertainty, provenance, admissibility, and the Alphonic limit.

**Generonic boundary and Transfactor.** The Generonic boundary marks the change in representational status through which a sensed interaction becomes a finite symbolic registration. A Transfactor denotes a functional element that performs or stabilises that sensing-to-symbol operation.

**Nexil.** The Nexil denotes a first-order finite symbolic registration generated from measurement at the relevant Alphonic context. It should not automatically be assigned to every downstream computational state.

**Finite Symbolic Dynamics.** FSD concerns the evolution, transformation, coupling, and relational organisation of finite symbolic registrations and their endogenous symbolic products.

**Functional Symbolic Trajectory.** An FST records an identifiable functional path through symbolic states and transformations while preserving the history required for interpretation.

**Dynexil.** The Dynexil remains a bounded local dynamical projection or delay-structured descriptor. Its role is to reopen local trajectory structure that a static state representation may compress.

### 14.3 Vocabulary Remains Subject to Reduction

The list above is not a protected ontology. It is a working grammar. Terms may later be merged, removed, or restricted if experimental applications show that they do not perform independent functional work.

This is not a weakness of the framework. It is a direct consequence of the functional discipline developed here. The language must remain capable of compression when a better generating commitment makes earlier subdivisions unnecessary.

## Chapter 15

# Functional Stability Across Transformations

### 15.1 The Stability Question

If a symbolic relation appears under one highly specific transformation but disappears under nearby admissible alternatives, its dependence upon the transformation should remain part of the result. If the relation survives many admissible transformations, its stability becomes more substantial.

Let

$$\Gamma^{(1)}, \Gamma^{(2)}, \dots, \Gamma^{(q)} \quad (15.1)$$

be different admissible representations derived from the same registered source. A relational feature  $\mathcal{K}$  may be considered more stable when

$$\mathcal{K}(\Gamma^{(1)}) \sim \mathcal{K}(\Gamma^{(2)}) \sim \dots \sim \mathcal{K}(\Gamma^{(q)}) \quad (15.2)$$

under a declared comparison rule.

The relation  $\sim$  is deliberately finite and operational. It need not imply classical identity.

### 15.2 Resolution Sensitivity

The base/depth framework provides a controlled way to test this idea. Let  $\Gamma_m$  denote the trajectory produced at numerical character depth  $m$ , all other conditions being held as closely as admissible. Then compare

$$\Gamma_6, \Gamma_7, \Gamma_8, \Gamma_9. \quad (15.3)$$

A trajectory feature that remains stable across these depth changes has greater evidence of symbolic robustness than one that moves substantially with every newly admitted character position.

A schematic resolution sensitivity measure may be written

$$S_R(m) = d(\Gamma_m, \Gamma_{m+1}), \quad (15.4)$$

where  $d$  is a declared finite trajectory comparison. No universal metric is assumed.

### 15.3 First Registered Distinction Time

For nonlinear trajectories, a finite alternative to infinitesimal separation can be defined. Let two admissible initial states be  $X_0$  and  $Y_0$ . If the working registration initially maps them to the same symbolic state, then they are not yet distinct within that representation.

Define the first registered distinction time

$$T_D = \min\{t : \mathcal{R}(X_t) \neq \mathcal{R}(Y_t)\}. \quad (15.5)$$

This quantity asks when two trajectories first become distinguishable under the declared symbolic system. It may later be compared with conventional Lyapunov-style analysis without requiring that the initial perturbation be infinitesimal.

## Chapter 16

# Conditions for Application to an External Dynamical System

### 16.1 A Readiness Condition

The framework is ready for a serious external application when it can be exposed to a new measurement domain without requiring a large prior expansion of terminology simply to make the application fit.

A provisional readiness condition is:

A finite symbolic framework is ready for application when its existing functional grammar can relate declared exogenous registrations, finite transformations, local trajectory descriptions, and downstream comparisons without first requiring uncontrolled lexical or formal growth.

The application may later reveal genuine missing operations. Those should be introduced because the function requires them, not because the language permits them.

### 16.2 The Application Chain

The intended general chain is

$$\text{exogenous interaction} \rightarrow \text{finite registration} \rightarrow \text{finite numerical representation} \rightarrow \text{endogenous transformation} \rightarrow \dots \quad (16.1)$$

Every arrow must remain operationally visible when it changes scale, uncertainty, numerical depth, neighbourhood structure, or admissibility.

### 16.3 No Hidden Infinite Precision

A later FSM computational experiment should not declare a nine-character numerical system while performing its true internal evolution in unrestricted higher precision and merely rounding the displayed result. That would place the finite rule at the reporting layer rather than inside the dynamics.

The working trajectory should therefore apply the admitted numerical registration at each declared transition:

$$N_i \xrightarrow{T_i} N_{i+1}^{\text{candidate}} \xrightarrow{\mathcal{R}_{10,9,e}} N_{i+1}. \quad (16.2)$$

If a higher-depth calculation is used as a comparison control, it must remain a separate finite trajectory with its own declared depth:

$$\Gamma_9 \quad \text{versus} \quad \Gamma_{12}, \quad (16.3)$$

for example. The higher-depth control is not treated as an exact hidden truth. It is another finite symbolic system against which the working system can be compared.

## 16.4 Spatial and Numerical Admissibility Together

For the proposed Navier–Stokes programme, two principal finite constraints will operate together. A spatial registration must satisfy

$$h \geq \alpha, \quad (16.4)$$

and a numerical registration will use

$$b = 10, \quad m = 9, \quad (16.5)$$

with explicit scale transitions.

This gives a combined state condition of the form

$$S_i^n = (\text{local registered variables}, \alpha, R_{10,9,e}, U_i^n, P_i^n, A_i^n). \quad (16.6)$$

The exact variable set belongs to the later fluid monograph. Part III establishes only the representational discipline under which those variables should evolve.

## Chapter 17

# Preparing the Navier–Stokes Stress Test

### 17.1 Why Navier–Stokes Is an Appropriate Next Application

The Navier–Stokes problem is a particularly valuable next test because its classical formulation begins from a continuum field and asks questions involving smoothness, differentiation, and possible singular behaviour. An FSM reformulation will instead begin from finite registered spatial relations and finite numerical states.

The goal will not be to claim a solution to the classical Millennium Prize problem. The goal will be to determine what problem remains when the continuum assumptions are replaced by the finite admissibility conditions developed through FSM, FSD, and the present numerical framework.

### 17.2 What Part III Adds to the Earlier Research Plan

An earlier outline of the fluid programme already imposed a spatial Alphonic floor. Part III now adds several necessary constraints.

First, the numerical evolution must itself be finite rather than being an unrestricted continuum computation sampled on a finite grid. Second, numerical scale changes must be recorded as changes in resolving power. Third, measurement-derived initial states must remain distinguishable from computationally evolved symbolic states. Fourth, Dynexil analysis should remain downstream and local rather than being confused with the update rule. Fifth, the growth of the explanatory vocabulary during the fluid study should itself be monitored.

These additions make the Navier–Stokes investigation a simultaneous test of fluid dynamics and of the functional economy of FSD.

### 17.3 A Two-Dimensional Stress Test of the Grammar

The later study can therefore be judged in two dimensions.

The first is physical/computational:

Does the finite spatial and numerical system produce stable, reproducible, and informative trajectories?  
(17.1)

The second is linguistic/framework-level:

Can the present grammar describe those trajectories without uncontrolled conceptual proliferation?  
(17.2)

A positive result in the second sense does not prove the physical model. It shows that the FSD language has transferred to a difficult new domain with functional economy. A negative result is equally informative because it identifies where the present grammar is under strain.

## Chapter 18

# Research Questions Arising from Part III

The present development suggests a focused rather than unlimited research programme.

The first question concerns functional termination. Can a local Functional Symbolic Trajectory be given operational stopping conditions that are stable under repeated use, while remaining reopenable by new exogenous measurement?

The second concerns framework growth. Can the changing complexity of a symbolic model be characterised well enough to distinguish productive elaboration from epicyclic proliferation without reducing all conceptual development to a single arbitrary metric?

The third concerns numerical representation. Can base, character depth, and scale be treated as independent declared properties of a finite numerical register in a way that remains computationally useful across arithmetic, simulation, and measurement?

The fourth concerns substrate independence. Which arithmetic relations remain functionally invariant across materially different implementations such as marks, beans, beads, mechanical counters, and electronic registers, and what must be preserved for that equivalence to hold?

The fifth concerns higher-order numerical representation. Is the existing language of Alphonic distinction, Nexil, finite symbolic state, and FST sufficient, or does repeated computational use demonstrate a genuine need for a specialised numerical Nexil or related term?

The sixth concerns local dynamical description. Does the Dynexil retain the same useful local function across measurement sequences, numerical trajectories, language, and physical simulation, or does its apparent portability disappear under detailed application?

The seventh concerns stability across resolution. Which macroscopic trajectory relations remain stable as numerical character depth and spatial resolution are varied within the admitted finite range?

The eighth concerns nonlinear separation. Can first registered distinction time, recurrence, delay reconstruction, and other finite trajectory measures provide useful alternatives or complements to infinitesimal nonlinear dynamical descriptions?

The ninth concerns the planned fluid application. Can the current grammar of FSM and FSD support a finite Navier–Stokes investigation without requiring an explosion of new concepts before the computation has even begun?

The final question is the broadest:

$$\boxed{\text{Can FSD remain functionally compact as the range of its applications grows?}} \quad (18.1)$$

## Chapter 19

# Provisional Principles of Part III

The argument developed in this monograph can be compressed into a small set of provisional principles. They are working constraints, not final axioms.

**Principle 1: Finite registration does not itself terminate symbolic growth.** A system composed entirely of finite symbols can still generate an indefinitely expanding explanatory trajectory.

**Principle 2: Symbolic growth is itself observable.** The development of a framework has order, provenance, recurrence, and local transformation structure and can therefore be examined as an FST.

**Principle 3: Function constrains elaboration.** A symbolic distinction earns stability when it performs repeatable functional work rather than merely increasing descriptive detail.

**Principle 4: Alphonic and functional termini are different.** The Alphonic terminus marks unavailable further distinction. The functional terminus marks locally unnecessary further elaboration.

**Principle 5: Functional termination is elastic.** A locally sufficient trajectory can be re-opened by new measurement, failed function, changed operating conditions, or newly reproducible distinction.

**Principle 6: Complexity growth is diagnostic.** Increasing symbolic burden with diminishing functional gain is a reason to inspect generating commitments rather than automatically add another partition.

**Principle 7: Base is not resolution.** Numerical resolution depends upon base, character depth, scale, and the registration rule. A finite base does not by itself imply either finite or infinite numerical resolution.

**Principle 8: Character depth is part of numerical provenance.** The number of admitted significant character positions must remain visible when numerical states are compared, transformed, or reported.

**Principle 9: Scale shifts change resolving power.** Moving a fixed-depth numerical register between scales changes its smallest available absolute positional distinction and is therefore part of the symbolic trajectory.

**Principle 10: Computation does not create measurement depth.** Additional digits generated by endogenous symbolic operations do not retroactively increase the resolution of the exogenous measurement from which the calculation began.

**Principle 11: Arithmetic is an operational symbolic trajectory.** Counting, carrying, shifting, grouping, and registering can be treated as finite state transitions instantiated through different physical substrates.

**Principle 12: Higher symbolic order is not higher physical dimension.** A multi-character numerical register is a higher-order relational symbolic construction. Its order does not by itself establish an additional measured physical dimension.

**Principle 13: Dynexil remains local.** The Dynexil is used as a bounded local dynamical descriptor and should not be expanded into a general update rule merely because a new application is encountered.

**Principle 14: Vocabulary remains reducible.** The terminology of FSM and FSD should be simplified when a smaller grammar preserves or improves function.

**Principle 15: External application is a stress test of the grammar.** A mature application should be allowed to reveal whether the existing language is sufficient before new terms are introduced.

## Chapter 20

# Conclusion: From Unlimited Description to Functional Sufficiency

Finite Symbolic Dynamics began by giving priority to the finite registered trajectory. It then expanded upstream toward the instruments and distinctions through which symbolic streams are produced. The present third part has examined a different boundary: not the boundary at which the first symbol appears, but the condition under which a symbolic trajectory may locally stop expanding.

The need for this boundary arose from the dynamics of the work itself. A finite framework can continue partitioning without contradiction. Each new partition may be intelligible. Yet intelligibility alone does not ensure functional economy. If a framework continually adds symbolic machinery in order to preserve its existing structure, the growth becomes information about the framework.

The neglected word *functional* therefore becomes central. A Functional Symbolic Trajectory is not merely a finite sequence that has been given a name. It is a finite historical pathway that performs a declared operation under identifiable conditions. Its growth is justified by function, and its local stopping condition is functional sufficiency.

This produces two complementary termini:

|                        |                   |                                |
|------------------------|-------------------|--------------------------------|
| measurement refinement | $\longrightarrow$ | $\alpha,$                      |
| symbolic elaboration   | $\longrightarrow$ | $\Gamma_{\text{functional}}^*$ |

(20.1)

The first is a hard boundary of distinction. The second is an elastic boundary of local sufficiency.

The numerical investigation makes the distinction concrete. A base is not an infinitely resolved numerical world. It is a finite symbolic alphabet with a positional and carry relation. A physically instantiated numerical system also possesses a finite character depth and an operating scale. For the present computational programme this is expressed through the base-10 nine-character register

$$R_{10,9,e}, \tag{20.2}$$

with least available positional distinction

$$\Delta_{10,9,e} = 10^{e-8}. \tag{20.3}$$

The absolute resolution changes with scale even though the number of character positions remains fixed.

Arithmetic can then be returned to its operational basis. Marks are added. Beans are grouped. Beads are moved. Registers change state. Carries and shifts propagate through finite positions. The material substrate changes, but stable functional relations can survive. The numerical trajectory is therefore not required to possess an infinitely resolved hidden value before the operation can take place.

This does not reject classical mathematics. It changes the order of explanation when mathematics is used as a representation of measured knowledge. A classical continuum or indefinitely extensible numeral may remain a powerful symbolic construction. Within FSM, the measurement-derived finite state and the endogenous construction should not be silently identified.

Part III therefore ends in a more stable but deliberately open position. The framework has not been completed. It has acquired a discipline for deciding when completion is locally sufficient. Its language can grow, but growth is now itself subject to examination. A new term, partition, or formal object is not justified merely because it can be introduced. It should show repeated function.

This creates the necessary condition for the next stage. The planned Navier–Stokes investigation can now be approached without first elaborating a specialised FSD vocabulary for fluids. The existing grammar can be exposed to the problem. Spatial distinction will be finite. Numerical character depth will be finite. Scale changes will be explicit. Exogenous registration will remain distinct from endogenous computation. Local Dynexils can be constructed downstream where they perform useful analytical work. The growth of the explanatory framework itself can be watched.

If the existing structure holds, the application will provide evidence that the recurrent conceptual basin of FSM and FSD is functionally stable. If it does not, the instability will no longer be treated automatically as a reason to add another epicycle. It will be evidence that one of the generating commitments requires reconsideration.

The global trajectory therefore remains open.

The local trajectory may stop.

And the condition for stopping is not that nothing more can be said, but that nothing more presently needs to be said for the declared function to hold.

# Working Notes for Transition to the Navier–Stokes Programme

The next monograph or research document can begin from the following held position.

The Navier–Stokes study should use a finite spatial representation with an explicit lower admissibility bound  $h \geq \alpha$ . It should not describe the finite grid merely as an approximation that conceptually tends toward zero spacing. Refinement should terminate operationally at the Alphonic boundary.

The numerical calculation should use an explicit finite register. The initial working condition is base 10 with nine significant numerical character positions at any represented scale. Every transition that changes scale must record the resulting change in absolute resolving power. Any higher-depth computation used for calibration or comparison should be treated as a separate finite trajectory rather than as an exact hidden reference.

Measurement-derived inputs should remain distinct from computationally evolved symbolic states. A measured initial or boundary condition may be represented through Nexils. The later endogenous cell states should not be relabelled as new measurements merely because the computation has advanced.

The fluid update rule should remain an explicit finite symbolic transformation. The Dynexil should be used only where a local delay-structured trajectory description is functionally useful. It should not be used as a synonym for the update rule, a spatial cell, or the global simulated state.

The study should compare trajectories across several admitted numerical depths and spatial scales to identify which macroscopic relations are stable and which are resolution dependent. It should record symbolic collision, scale shift, saturation, overflow, and any demand for sub-Alphonic refinement as first-class events in the trajectory.

Finally, the vocabulary of the study should be monitored. New terms should be introduced only where the existing functional grammar cannot describe a repeated and important operation without ambiguity. The Navier–Stokes programme is therefore not only an application of FSD. It is the first substantial external stress test of whether the language developed in Parts I–III has reached a useful local functional terminus.